MARKING SCHEME

B. MATHEMATICS ARUSHA REGION TERMINAL EXAMINATION.
AUGUST 2020. FORM THREE.

1 (a) (i) 784.37 (Hundredths) — (01)

(ii) 800 (Hundreds) — (01)

(iii) 784.4 (4 S. figure). — (01)

(b) Let the first number be X.
2nd number be Y.

$3/x \; \frac{1}{3}(x+y) = 50 \times 3$

$5/x \; \frac{1}{5}(x-y) = 2 \times 5$ (01)

$$\begin{cases} x+y = 150 \\ x-y = 10 \end{cases}$$

$$\frac{0+2y}{2} = \frac{140}{2}$$

$y = 70$ (01)

Substitute $y = 70$ into equation (1)

$x+y = 150$

$x+70 = 150$

$x = 150-70$

$x = 80$ (01)

First number is 80 and second number is 70.

—01)

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2 (a) find the value of x if $\left(\frac{2}{3}\right)^{2x-3} = \left(\frac{243}{32}\right)$

$\left(\frac{2}{3}\right)^{2x-3} = \frac{3^5}{2^5}$

$\left(\frac{2}{3}\right)^{2x-3} = \left(\frac{3}{2}\right)^5$

$\left(\frac{2}{3}\right)^{2x-3} = \left(\frac{2}{3}\right)^{-5}$ — 01

Equal base equal exponent

$\left(\frac{2}{3}\right)^{2x-3} = \left(\frac{2}{3}\right)^{-5}$ — 01

$2x-3 = -5$

$2x-3+3 = -5+3$

$\frac{2x}{2} = \frac{-2}{2}$

$x = -1$

∴ The value of $x = -1$ — 01

(b) Given $\log x = \bar{5}.3567$. find $\log \sqrt[3]{x}$

from $\log \sqrt[n]{x} = \log x^{1/n}$. — 01

$\log \sqrt[3]{x} = \log x^{1/3}$

$= \frac{1}{3} \log x$ but $\log x = \bar{5}.3567$.

$= \frac{1}{3} \times \bar{5}.3567$.

$= \frac{\bar{5}.3567}{3}$ — 01

$\frac{\bar{5} + 0.3567}{3}$

$\frac{\bar{5} + \bar{1} + 1.3567}{3}$

$\frac{\bar{6} + 1.3567}{3}$

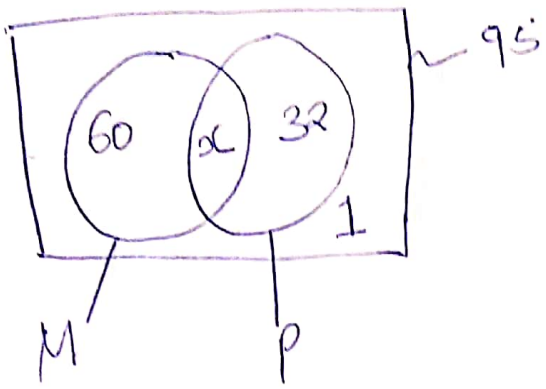
$\bar{2} + 0.4522$

$\bar{2}.4522$ — 01

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3.

(a)



let Mathematics be M
physics be P.

from

$n(M \cup P) = n(M) + n(P) - n(M \cap P)$

but $n(M \cup P) = M - n(M \cup P)'$

$= 95 - 1.$

$= \underline{94}$ — (01)

$60 + x + 32 = 94$

$92 + x = 94$

$x = 94 - 92$

$\underline{x = 2}$ — (01)

(i) Students study Mathematics is $60 + x$

$60 + 2$

$= \underline{62}$ — (01)

(ii) Students study both subject is $\underline{2}$

(b) $n(A \cap B) = 10 + x$

$n(A) = 5x$

$n(B) = 2x$

$n(A \cup B) = 50$ find the value of x.

from $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ — (01)

$50 = 5x + 2x - (10 + x)$

$50 = 7x - 10 - x$

$50 = 6x - 10$

$6x = 50 + 10$

$\frac{6x}{6} = \frac{60}{6}$ $\underline{x = 10}$ — (02)

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4 (c) X-intercept = 2 $\Rightarrow$ (2, 0)

y-intercept = 4 $\Rightarrow$ (0, 4)

find the equation in the form of $y = mx + c$

$$\text{Slope} = \frac{\Delta y}{\Delta x}$$

$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

$$M = \frac{4 - 0}{0 - 2}$$

$$M = \frac{4}{-2}$$

$$M = -2$$ — 01

Let other point (x, y) to any point let $(0, 4)$ when the Slope is -2.

$$M = \frac{\Delta y}{\Delta x}$$ — 01

$$\frac{-2}{1} = \frac{y - 4}{x - 0}$$

$$y - 4 = -2(x - 0)$$

$$y - 4 = -2x$$

$$y = -2x + 4$$

∴ The equation is $y = -2x + 4$ — 01

(b.) from

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$ Where $\begin{pmatrix} x' \\ y' \end{pmatrix}$ = image — 01

$\begin{pmatrix} x \\ y \end{pmatrix}$ = object

$\begin{pmatrix} a \\ b \end{pmatrix}$ = translation vector.

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$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = ?$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{pmatrix} 3 \\ -5 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ -5 \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \end{pmatrix} \quad \text{— 01}$$

Now

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = ? \quad \text{when} \quad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2 \\ 7 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 5 \\ -6 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$$

$$= \begin{pmatrix} 2 \\ 7 \end{pmatrix} + \begin{pmatrix} 5 \\ -6 \end{pmatrix}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix} \quad \text{— 01}$$

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5(a) Exterior angle + interior angle = 180

$$x + (x+120) = 180^\circ$$

$$2x + 120 = 180^\circ$$

$$2x = 180 - 120$$

$$\frac{2x}{2} = \frac{60}{2}$$

$$x = 30^\circ$$ (01)

But the measures of exterior angle is given by

$$\frac{360}{n} = 30^\circ$$ (01)

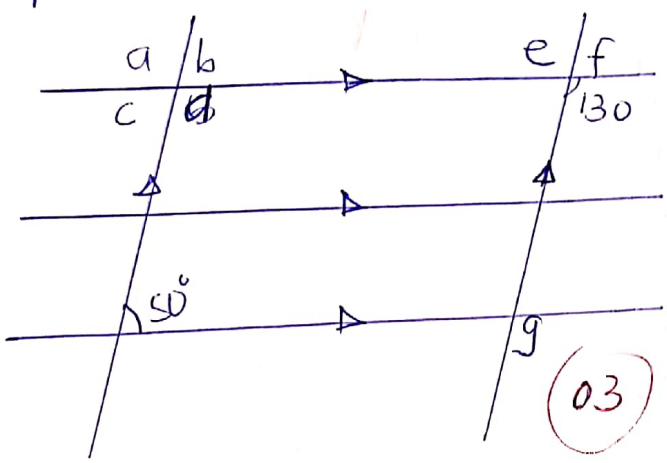
$$n = \frac{360}{30}$$

$$n = 12$$ (01)

The number of sides of a polygon are 12

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5 (b) From the illustration given



03

$a = d$ (vertically opposite)
$d = 130°$ (corresponding)
hence $a = 130°$
$b = c$ (vertically opposite)
$b = 50°$ (corresponding)
$c = 50°$
$e = 130°$ (vertically opposite)
$g = 130°$ (corresponding)
$f = 50°$ (vertically opposite).

6 (c) U$ 1200 x 3 = 3600 U$

(i) from 1 U$ = Tsh 2280
3600 U$ = x

$$x = \frac{\text{Tsh } 2280 \times \text{U\$ } 3600}{1 \text{ U\$}}$$

x = Tsh 8208000/=

The amount was Tsh 8208000/= 1½

(ii) From 1 Ksh = 22 Tsh
y = Tsh 8208000/-

$$y = \frac{1 \text{ Ksh} \times \text{Tsh } 8208000}{22 \text{ Tsh}}$$

y = 373090.91

Amount in ksh was 373090.91 — 1½

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6 (b) $R \propto \frac{L}{A}$ — (01)

$R = \frac{KL}{A}$

$\frac{KL}{L} = \frac{RA}{L}$

$K = \frac{RA}{L}$ but $R_1 = 8750\Omega$

$A_1 = 4M^2$

$L_1 = 2CM \rightarrow$

$L_2 = 36CM$

$A_2 = 3M^2.$

$K = \frac{R_1A_1}{L_1} = \frac{R_2A_2}{L_2}.$ $R_2 =$? — (01)

$\frac{R_2A_2}{A_2} = \frac{R_1A_1}{L_1} \cdot \frac{L_2}{A_2}.$

$R_2 = \frac{8750\Omega \times 4M^2 \times 36CM}{2CM \times 3M^2}$

$R_2 = 210000\Omega$ — (01)

OR

$K = \frac{RA}{L}$

$K = \frac{8750 \times 4M^2}{0.02m}$ — (01)

$K = 1750,000$

NOW

$K = \frac{R_2A_2}{L_2}$

$1750000 = \frac{R_2 \times 3}{0.36}$

$R_2 = \frac{1750000 \times 0.36}{3}$

$R_2 = 210,000\Omega$ — (02)

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7 (a) let A be V.A.T and
X be price before V.A.T

So $A + X = 133800$ --- (i) (01)

But $A = 30\%$

$\frac{30}{100}$ of $x = \frac{3}{10}x$

Substitute in (i) above

$\frac{3}{10}x + \frac{x}{1} = 133800$ — (01)

$10 \times \frac{3x + 10x}{10} = 133800 \times 10$

$\frac{13x}{13} = \frac{1338000}{13}$

$x = 102923.10$

So the price before V.A.T was

Tsh 102923.10 (01)

(b) $A : B = 3 : 4$

$B : C = 5 : 7$

(i) $A : C = \frac{A}{B} \times \frac{B}{C}$

$A : C = \frac{3}{4} \times \frac{5}{7}$

$A : C = 15 : 28$

$A : B = 5 | 3 : 4$

$B : C = 4 | 5 : 7$

$A : B = 15 : 20$

$B : C = 20 : 28$

$A : C = 15 : 28$ — 1½

$A : B : C = 15 : 20 : 28$ 1½

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8. (a) 4, x, y and 22 are in A.P.

(i) Value of x and y.

for A.P $x-4 = y-x = 22-y$.

$x-4 = y-x$ — 01½

$2x-y = 4$ --- (i)

$y-x = 22-y$

$2y-x = 22$ --- (ii).

$$\begin{cases} 2x-y=4 & -(i) \\ -x+2y=22 & -(ii) \end{cases}$$

$$+\begin{cases} 2x-y=4 \\ -2x+4y=44 \end{cases}$$

$0+3y = 48$

$\frac{3y}{3} = \frac{48}{3}$

$y = 16$ — 01½

Substitute $y=16$ into eqn (i)

$2x-y=4$

$2x-16=4$

$2x = 4+16$

$\frac{2x}{2} = \frac{20}{2}$

$x = 10$ — 01½

∴ The value of $x=10$ and $y=16$

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8 (ii) The sum of the first 16th term of A.P.

4, 10, 16, 22

$A_1 = 4, \quad A_2 = 10, \quad A_3 = 16$

$d = A_2 - A_1 = A_3 - A_2$

$= 10 - 4 = 16 - 10$

$d = 6$ — 01/2

$S_n = \frac{n}{2}(2A_1 + (n-1)d)$.

$S_{16} = \frac{16}{2}(2 \times 4 + (16-1)6.)$

$S_{16} = 8(8 + (15 \times 6))$.

$= 8(8 + 90)$

$= 8 \times 98$

$S_{16} = 784$ — 01

∴ The sum of the first 16th term of an A.P is 784

(b) $G_3 + G_4 = 12$ - - - (i)

$G_4 + G_5 = 36$. - - - (ii)

Take eqn (ii) divide by eqn (i)

$\frac{G_4 + G_5}{G_3 + G_4} = \frac{36}{12}$.

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from $G_n = Gr^{n-1}$

$$\frac{G_4 + G_5}{G_3 + G_4} = \frac{36}{12}$$

$$\frac{Gr^3 + Gr^4}{Gr^2 + Gr^3} = 3$$

$$\frac{Gr^3(1+r)}{Gr^2(1+r)} = 3$$

$$r^{3-2} = 3$$

$$r = 3$$

(i). Common ratio is 3 — 1½

$$G_4 + G_5 = 36$$

$$Gr^3 + Gr^4 = 36$$

$$Gr^3(1+r) = 36$$

$$G1 \times 3^3(1+3) = 36$$

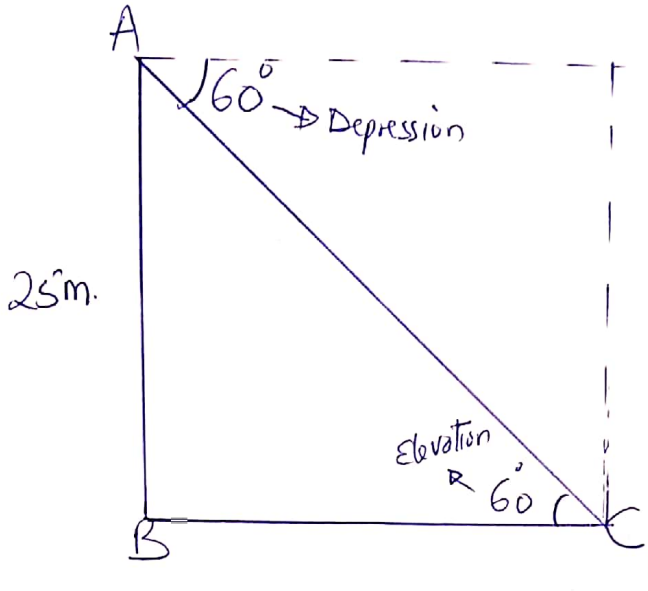
$$\frac{G \times 27 \times 4}{27 \times 4} = \frac{36}{27 \times 4,}$$

$$G_1 = \frac{3}{9} = \frac{1}{3}$$

(ii) The first term is $\frac{1}{3}$ — 1½

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9 (c) Consider



01

$$\sin\theta = \frac{\text{opposite}}{\text{Hypotenus}}$$

$$\sin 60^\circ = \frac{25}{\text{Boat distance}}$$

$$\therefore \text{Boat distance} = \frac{25}{\sin 60^\circ}$$

$$= \frac{25}{0.8660}$$

$$= 28.87$$ 01

The boat is at a distance of 28.87m from the top of a cliff.

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9 (b) find the value of $\frac{\sin 60}{\cos 60 \sin 30}$.

but $\sin 60 = \sqrt{3}/2$

$\cos 60 = 1/2$

$\sin 30 = 1/2$. +0½

$$\frac{\sin 60}{\cos 60 \sin 30} = \frac{\sqrt{3}/2}{1/2 \times 1/2}$$

$$= \sqrt{3}/2 \div 1/4.$$

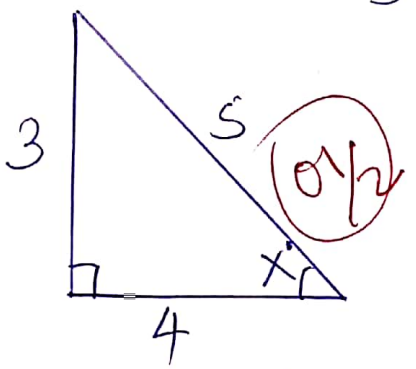
$$= \sqrt{3}/2 \times 4/1$$

$$= \frac{4\sqrt{3}}{2}$$ 1½

$$\frac{\sin 60}{\cos 60 \sin 30} = 2\sqrt{3}$$

(c) $\sin x = 0.6 \Rightarrow \frac{6}{10} = \frac{3}{5}$.

$\sin x = \frac{\text{opposite}}{\text{hypotenuse}}$.



0½

By pythagoras theorem

$a^2 + b^2 = c^2$

$3^2 + b^2 = 5^2$

$b^2 = 5^2 - 3^2$

$b^2 = (5+3)(5-3)$

8×2

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9 $\sqrt{b^2} = \sqrt{16}$

SOTOCA
HAH

$b = 4$ — 01/2

find $\dfrac{\sin x + \cos x}{\tan x}$.

$\sin x = 3/5$

$\cos x = 4/5$

$\tan x = 3/4$.

$$\frac{\sin x + \cos x}{\tan x} = \frac{3/5 + 4/5}{3/4}$$

$$= \frac{3+4}{5} \div 3/4$$

$$= 7/5 \div 3/4$$

$$= 7/5 \times 4/3$$

$$= \frac{28}{15}$$

$\dfrac{\sin x + \cos x}{\tan x} = 1.866$ — 01

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10 (a) Required to solve $ax^2+bx+c=0$ by completing the square

$$ax^2+bx+c=0$$

$$x^2+\frac{b}{a}x+\frac{c}{a}=0$$

$$x^2+\frac{b}{a}x=-\frac{c}{a}$$

$$x^2+\frac{b}{a}x+\left(\frac{b}{2a}\right)^2=\frac{-c}{a}+\frac{b^2}{4a^2}$$

$$\sqrt{\left(x+\frac{b}{2a}\right)^2}=\sqrt{\frac{b^2-4ac}{4a^2}}$$

$$x+\frac{b}{2a}=\pm\frac{\sqrt{b^2-4ac}}{2a}.$$

$$x=-\frac{b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}.$$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$$

02

(b) Required to solve $x^2-3x+2=0$ by quadratic formula.

$a=1$
$b=-3$
$c=2.$

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}.$$

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$$x = \frac{3 \pm \sqrt{9 - 4 \times 1 \times 2}}{2 \times 1}$$

$$= \frac{3 \pm \sqrt{9 - 8}}{2}$$

$$x = \frac{3 \pm \sqrt{1}}{2}$$

$$x = \frac{3 \pm 1}{2}$$

$$x = 4/2 \text{ or } 2/2.$$

$$x = 2 \text{ or } 1$$

$\therefore X = 2$ or $X = 1$ 02

10 (c). $\left(a - \frac{1}{a}\right)^2 = 10$ required to find the value of $a^2 + \frac{1}{a^2}$.

$$a^2 - 2 + \frac{1}{a^2} = 10$$

$$a^2 + \frac{1}{a^2} = 10 + 2$$

$$a^2 + \frac{1}{a^2} = 12$$ 02

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SECTION B.

11.

Class interval (X1000)	class marks (X) (X1000)	f	d = X − A	fd
20 − 29	24.5	2	−30	−60
30 − 39	34.5	5	−20	−100
40 − 49	44.5	7	−10	−70
50 − 59	54.5	11	0	0
60 − 69	64.5	9	10	90
70 − 79	74.5	3	20	60
80 − 89	84.5	3	30	90

01 01 $\Sigma f = 40$ 01 01 $\Sigma fd = 10$

$A = 54.5$

(a) $\bar{X} = A + \frac{\Sigma fd}{\Sigma f}$ 01

$= 54.5 \times 1000 + \frac{10 \times 1000}{40}$

$= 54500 + 250$

$= 54750$

∴ The mean wage is Tsh 54750 01

(b) Mode:

Modal class = 50 − 59

$L = 50 - 0.5$

$= 49.5 \times 1000$

$= 49500$

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11(b)

$t_1 = 11-7 = 3$

$t_2 = 11-9 = 2$

$i = 10$

$$\text{Mode} = L + \left(\frac{t_1}{t_1+t_2}\right) i \quad (01)$$

$$= 49.5 \times 1000 + \left(\frac{3}{3+2}\right) 10 \times 1000$$

$$= 49500 + \frac{3}{5} \times 10000$$

$$= 49500 + 6000$$

$$= 55500$$

$\therefore$ Mode = Tsh 55500 (01)

(c) Median:

Median class = 50 – 59, $L = 49.5$

$N = 40$

$C.f_b = 14$

$$\text{Median} = L + \left(\frac{\frac{N}{2} - C.f_b}{f_m}\right) i \quad (01)$$

$$= 49.5 \times 1000 + \left(\frac{\frac{40}{2} - 14}{11}\right) \times 10 \times 1000$$

$$= 49500 + \frac{6}{11} \times 10,000$$

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11(c). $= 49500 + 5454.55$

$= 54954.55$

$\therefore$ Median = Tsh 54954.55 — 01

12(a) Given: $R\{(x,y): x+y \leq 1 \text{ and } y \leq x-1\}$

$R^{-1}\{(x,y): y+x \leq 1 \text{ and } x \leq y-1\}$ 1/2

$R^{-1}\{(x,y): y \leq 1-x \text{ and } x+1 \leq y\}$

$\therefore R^{-1}\{(x,y): y \leq 1-x \text{ and } y \geq x+1\}$

Table of values for $y = 1-x$

x	0	1
y	1	0

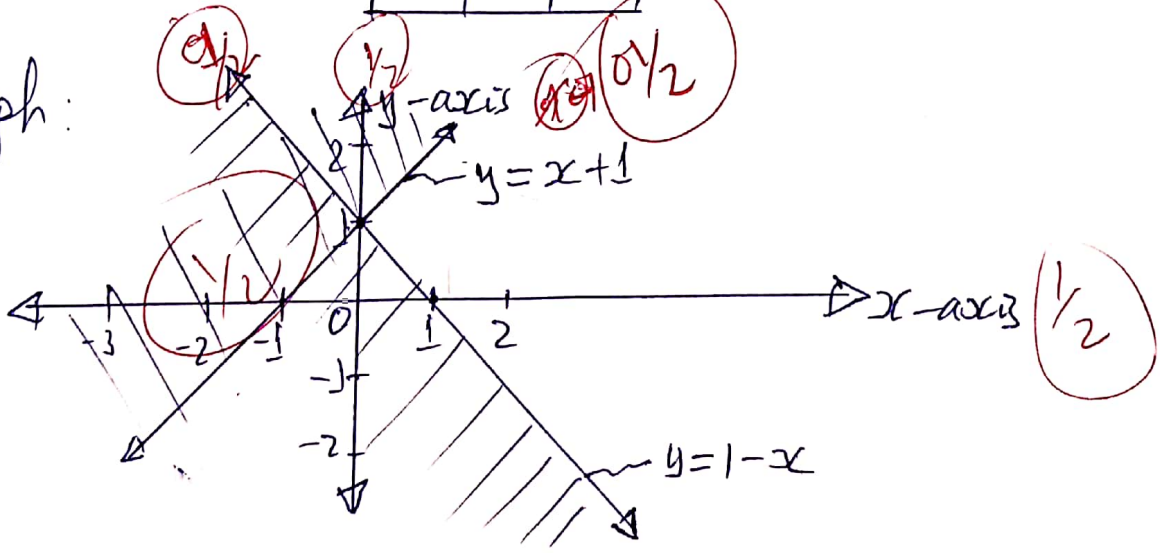
1/2

Table of values for $y = x+1$

x	0	-1
y	1	0

1/2

Graph:



01/2 1/2 01/2 1/2

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12(a) Domain of $R^{-1} = \{x : x \leq 0\}$ 01/2

Range of $R^{-1} = \{y : y \in R\}$ 01/2

(b)(i) $R = \{(x, y) : y = 2x^2 - 5x + 3\}$

Domain $= \{x : x \in R\}$ 01

Range $= \{y : y \geq \frac{4ac - b^2}{4a}\}$ 1/2

$= \{y : y \geq \frac{4(2)(3) - (-5)^2}{4(2)}\}$

$= \{y : y \geq \frac{24 - 25}{8}\}$

$= \{y : y \geq -\frac{1}{8}\}$

$\therefore$ Range $= \{y : y \geq -0.125\}$ 01

(ii) Table of values for $y = -x^2$

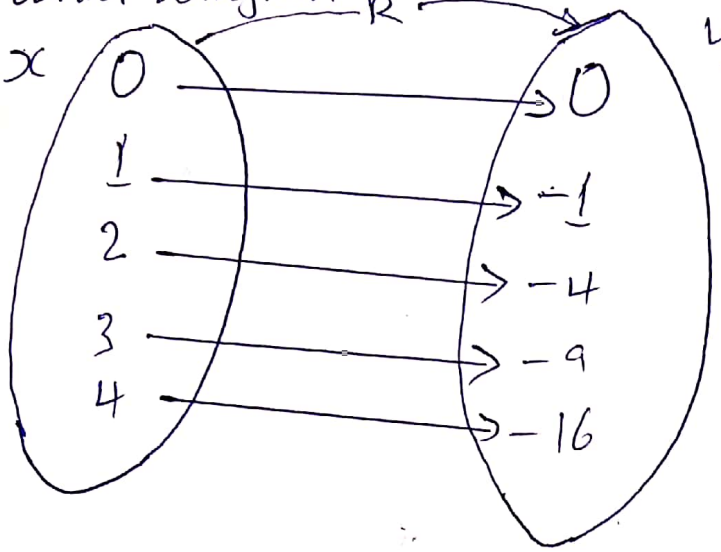
$x = A$	0	1	2	3	4
y	0	-1	-4	-9	-16

01/2

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12(b)(ii)

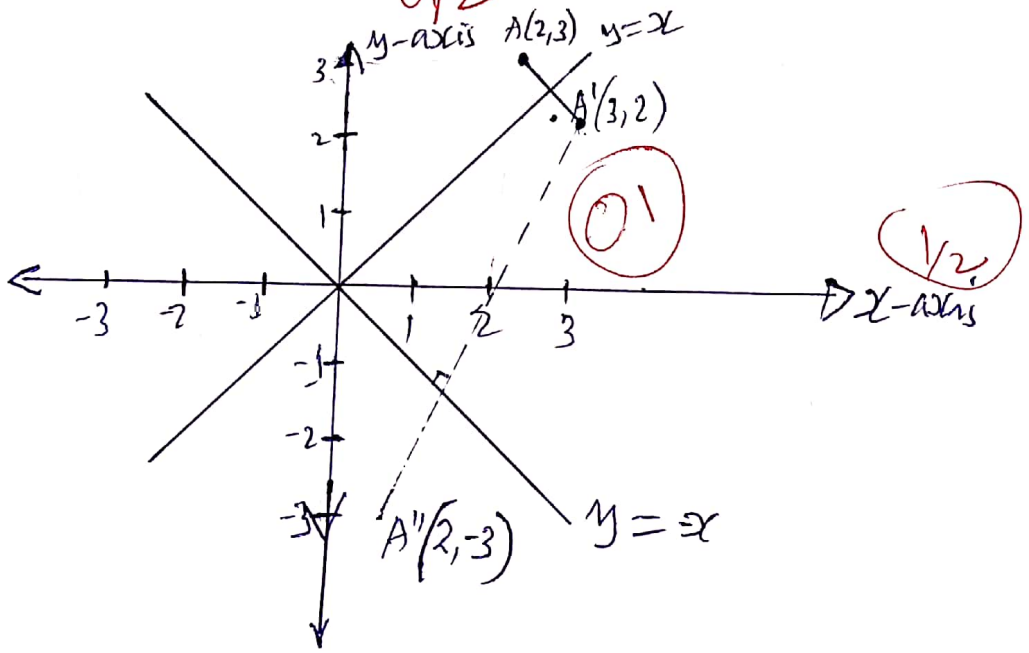
Pictorial diagram R



02

13(a)

0½



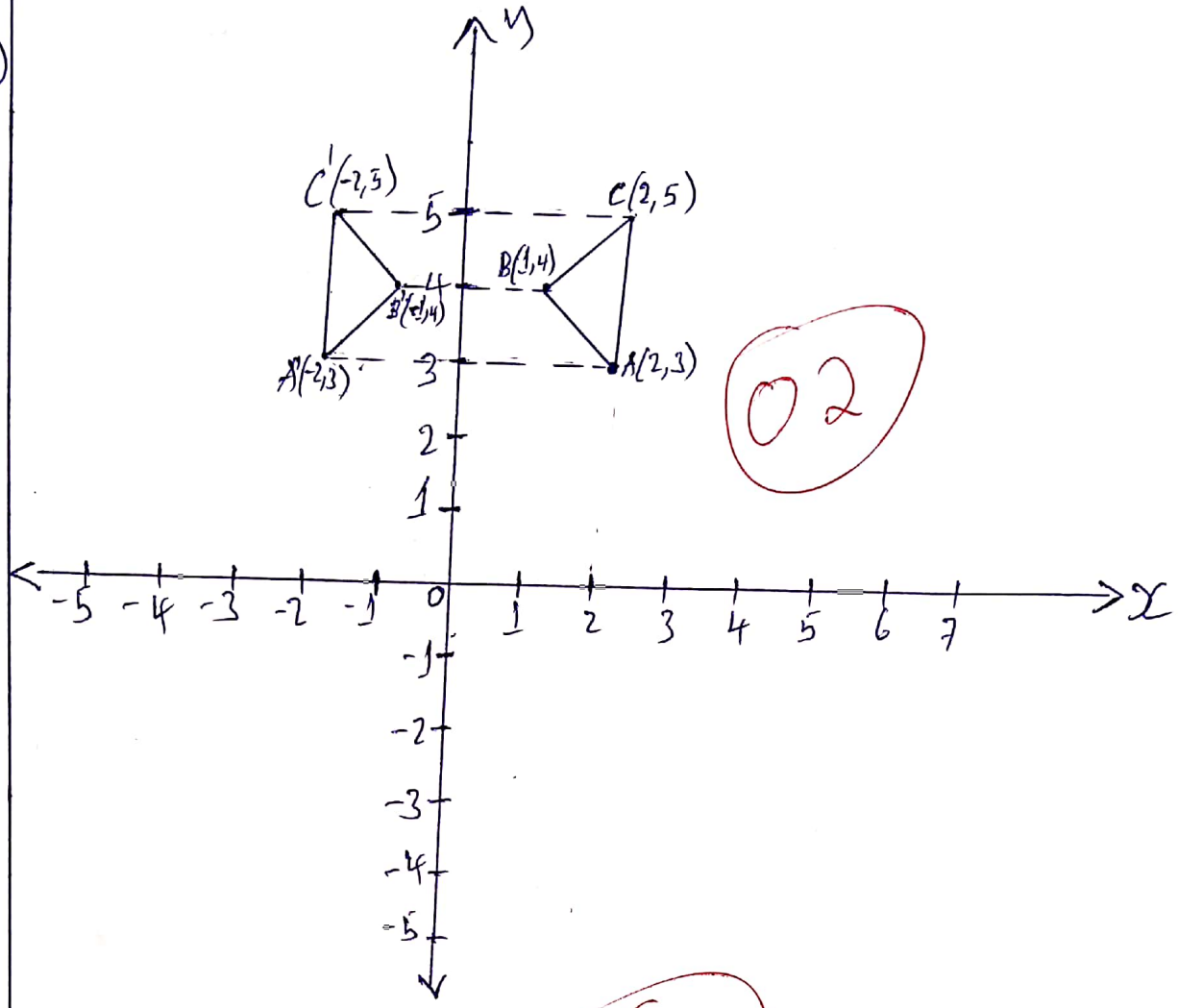
01

½

$\therefore A'' = (2, -3)$

01

13(b)



02

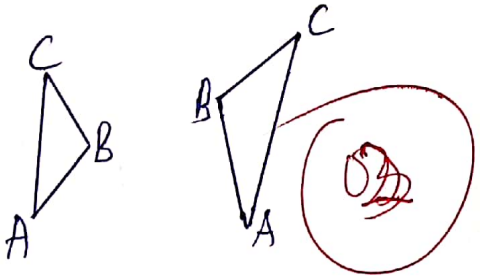
(i) $M_y(2,3) = (-2,3)$, $M_y(1,4) = (-1,4)$, $M_y(2,5) = (-2,5)$

01

∴ The coordinates of the image are $A'(-2,3)$, $B'(-1,4)$ and $C'(-2,5)$.

02

(ii)



02

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14

14(a) (i) $f(x) = 2x^2 + 3x + 1$

$= 2\left(x^2 + \frac{3}{2}x + \frac{1}{2}\right)$

$= 2\left(x^2 + \frac{3}{2}x + \left(\frac{3/2}{2}\right)^2 + \frac{1}{2} - \left(\frac{3/2}{2}\right)^2\right)$

$= 2\left(x^2 + \frac{3}{2}x + \left(\frac{3}{4}\right)^2 + \frac{1}{2} - \left(\frac{3}{4}\right)^2\right)$

$= 2\left(x + \frac{3}{4}\right)^2 + \frac{1}{2} - \frac{9}{16}$

$\therefore f(x) = 2\left(x + \frac{3}{4}\right)^2 - \frac{1}{16}$ — (01)

Turning point:

$x + \frac{3}{4} = 0$

$x = -\frac{3}{4}, \quad y = -\frac{1}{16}$

$\therefore (x, y) = \left(-\frac{3}{4}, -\frac{1}{16}\right)$ is the turning point.

— (01)

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14 (a) (i) $f(x) = ax + b$

$f(2) = a(2) + b$

$6 = 2a + b$ - - - (i)

$f(3) = a(3) + b$

$4 = 3a + b$ - - - (ii)

By solving eqns (i) and (ii) using elimination method

$$-\begin{cases} 2a + b = 6 \\ 3a + b = 4 \end{cases}$$ (01)

$-a = 2$

$a = -2$

From eqn (i), $2a + b = 6$

$2(-2) + b = 6$

$-4 + b = 6$

$b = 10$

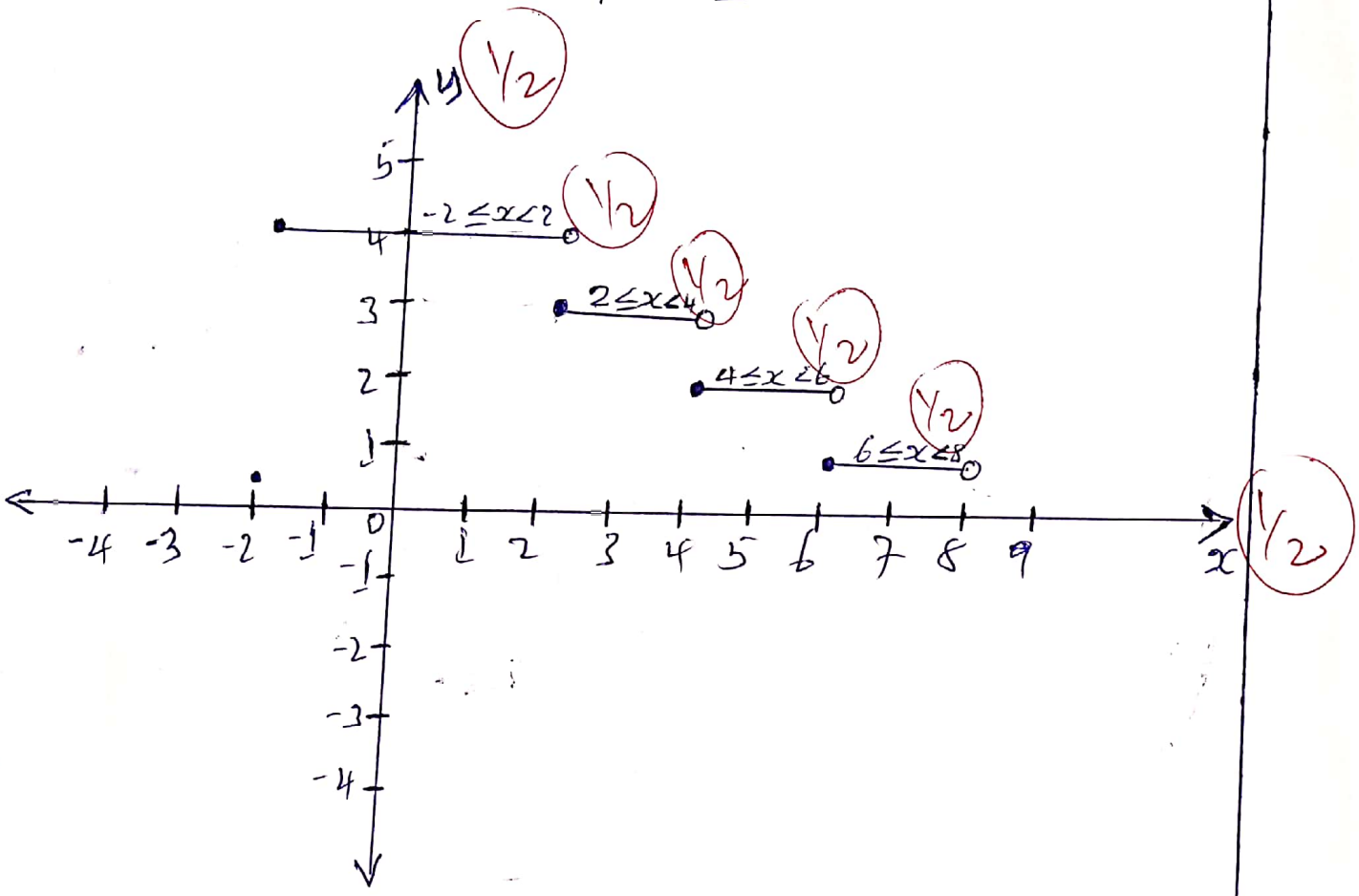
$\therefore a = -2$ and $b = 10$ (01)

The function is $f(x) = -2x + 10$ — (01)

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14(b) A graph of $f(x) = \begin{cases} 4 \text{ if } -2 \leq x < 2 \\ 3 \text{ if } 2 \leq x < 4 \\ 2 \text{ if } 4 \leq x < 6 \\ 1 \text{ if } 6 \leq x < 8 \end{cases}$

(i)



(ii) Domain $= \{x : -2 \leq x < 8\}$ (01)

Range $= \{4, 3, 2, 1\}$ (01)

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